

Deep Inelastic Scattering Revisited

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Abstract

Discuss Yadism [\[1\]](#).

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Chapter 1

The basics

1.1 Introduction

1.1.1 Personal introduction

- Ich + paper + yad

1.1.2 Goals and non-goals

Let us quickly review the goals and also the non-goals:

Goals:

- Make a meaningful comparison between the theoretical predictions and the experimental measurement of a fully inclusive DIS cross-section.
- Review pQCD features and exemplify them using DIS

Non-goals:

- We only consider pQCD in the collinear framework here. However, because DIS is such a powerful experiment, which offers a clean access to the inner structure of the protons it can be studied using other frameworks as well, such as Color Glass Condensates [2,3] → ask H. Mäntysaari or T. Lappi
- We only consider fully inclusive cross-sections here (to be defined in Section 1.2 and with a tiny exception to be discussed in Section 2.4). In particular, we will

Standard Model of Elementary Particles

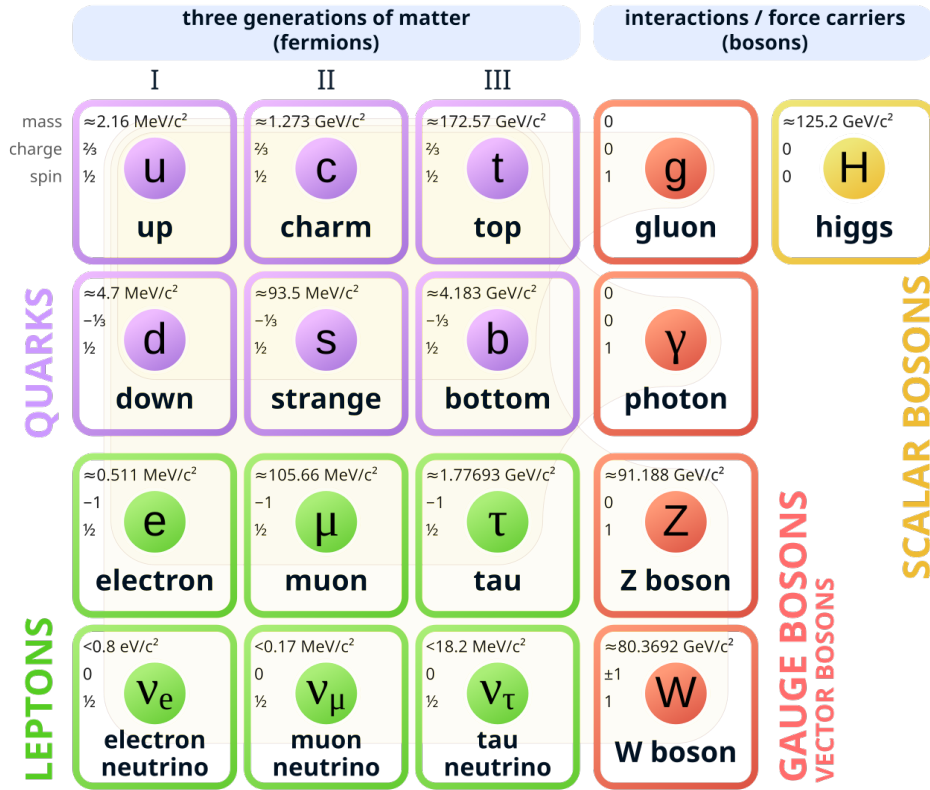


Figure 1.1: The Standard Model of Elementary Particles - taken from [7]

not consider any more complicated final states, such as, e.g. single jet or di-jet production [4].

- We only consider pQCD in the collinear framework here. While, Monte Carlo Event generators, such as, e.g. Pythia [5], are rooted in collinear pQCD they eventually go much beyond and we do not consider their details here [6] → ask I. Helenius.

1.1.3 Requirements

We need some basic knowledge from particle physics and, in particular, you need to be familiar with the Standard Model overview shown in Fig. 1.1. We will discuss in this course each and everyone of these particles, with the exception of the three heaviest in each category: the heaviest quark, the top, the heaviest lepton, the tau, and the heaviest boson, the Higgs. They will work, in principle mostly like their lighter brothers

and sisters, but in practice are basically always neglected. We give a more thorough justification to discard them when we get to the actual numbers. All other particles will explicitly appear in this course and you need to know their charges - see also Fig. 1.1.

Summary forces, particles, and charges

- The carrier of the strong force is the gluon
- Quarks have a strong charge, leptons don't
- The carriers of the electro-weak force are the photon, the Z boson, and the W boson
- Quarks and leptons have both electro-magnetic and weak charges

We also need some basic knowledge from Quantum Field Theory.

1.1.4 The very basics

- D I S
- factorization theorem
- LO (+ NLOq + NLOg)
- IC

1.2 Setting the stage

- x, Q2
- $L_{\mu\nu}W^{\mu\nu}$

We need the Riemann Zeta function $\zeta(n)$ below, which is defined at integer values n by

$$\zeta_n = \zeta(n) = \sum_{j=1}^{\infty} \frac{1}{j^n} \quad (1.1)$$

1.3 Distributions

As we have seen the coefficient functions may contain distributions. For example the quark coefficient function at LO is just a Dirac delta distribution:

$$C_q^{(0)}(z) = \delta(1 - z) . \quad (1.2)$$

To better compare with our next step, let's repeat the definition of the Dirac delta distribution:

$$\int_0^1 dz \delta(1 - z) f(z) = f(1) . \quad (1.3)$$

However, this is only the simplest case, which appears at LO, and at higher order we get more complicated distributions. For example the quark coefficient function at NLO is given by

$$C_q^{(1)}(z) = C_F \left[4 \left(\frac{\ln(1 - z)}{1 - z} \right)_+ - 3 \left(\frac{1}{1 - z} \right)_+ - (9 + 4\zeta_2) \delta(1 - z) - 2(1 + z) \ln \left(\frac{1 - z}{z} \right) - 4 \frac{\ln(z)}{1 - z} + 6 + 4z \right] \quad (1.4)$$

FH! where the first line only contains distributions. **FH: Where to define NLOq?** Here, we have introduced the usual $+$ -distributions, which are defined via

$$\int_0^1 dz a(z)_+ f(z) = \int_0^1 dz a(z) (f(z) - f(1)) . \quad (1.5)$$

We review in a moment again on how these distributions appear, but for the moment we turn to more practical matters as the goal of our course is to get an actual comparison to experimental data. The problem is: distributions are (complicate) math objects, defined only via integral relations, Eqs. (1.3) and (1.5), which are unsuitable for numeric evaluation.

Chapter 2

DIS in real life

2.1 More particles

- γ/Z
- leptons
- hadrons: $p+d+A$; nucl. PDF
- experimental overview

2.2 DGLAP

- resummation
- Mellin

2.3 Scale variations

- scheme A/B/C
- MHOU
- covariance matrix

2.4 Heavy Quarks

- F_2^c
- FONLL

2.5 γ_5

- helicity
- HVBM + Larin

Chapter 3

More advanced topics

3.1 Positivity

- DIS vs. MSbar

3.2 Higher orders

- IHOU
- N3LO
- channels
- loops and legs

3.3 QED

- Qiu et al.

3.4 Valo

- $C_\gamma \rightarrow k$

Bibliography

- [1] A. Candido, F. Hekhorn, G. Magni, T. R. Rabemananjara, and R. Stegeman, Eur. Phys. J. C **84**, 698 (2024), 2401.15187.
- [2] M. Bertilsson, T. Lappi, H. Mäntysaari, and X.-B. Tong, (2026), 2601.07302.
- [3] C. Casuga and H. Mäntysaari, (2026), 2604.22332.
- [4] H1, V. Andreev *et al.*, Eur. Phys. J. C **75**, 65 (2015), 1406.4709.
- [5] C. Bierlich *et al.*, SciPost Phys. Codeb. **2022**, 8 (2022), 2203.11601.
- [6] I. Helenius, J. O. Laulainen, and C. T. Preuss, (2026), 2605.00502.
- [7] Wikipedia, File:Standard Model of Elementary Particles.svg, online, 2026.